Garbage Collectors

Every Hadamard test throws away half its data. We recycle it.

Team 8

TBD-NAME TBD-NAME TBD-NAME

Qiskit Hackathon 2026 — Topic 5

after Faehrmann, Eisert & Kueng, *PRL* **135**, 150603 (2025)

14 August 2026

Outline

The garbage state

Does it work?

What each choice buys

Real hardware, and the price

Reproduce it, and where it breaks

A tag like [R034] is not decoration — it is a **row number** in our public evidence ledger, `RESULTS.md`: 44 rows, one per reported number, each with the exact command that produced it. Full repo via the QR code on the last slide.

The garbage state

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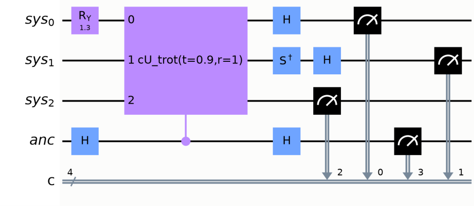
Every Hadamard test throws away half its data

The standard Hadamard test measures **one ancilla qubit** and **discards the system register**.

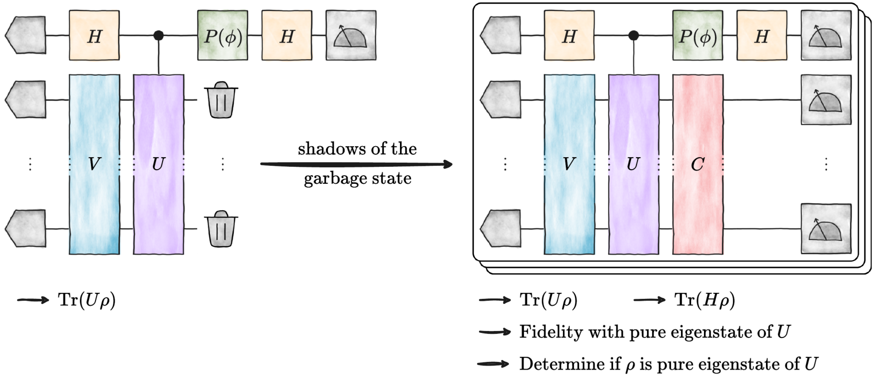
Measure that "garbage" in *random bases* instead and it becomes a **classical shadow** — the same shots now also carry the system's observables.

We implement this end to end, validate every estimate against exact references, *quantify what it costs*, and characterise where it breaks.



3 system qubits + 1 ancilla

The protocol, in the authors' own picture

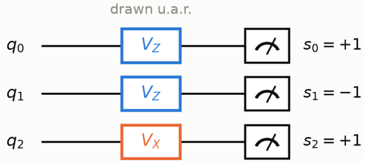


Left: the standard test — V prepares, c-U evolves, $P(\phi)$ picks the quadrature, and the system register goes **in the bin**. **Right:** append a random Clifford C and measure it instead: **same circuit, same shots**, now also $\mathrm{Tr}(H\rho)$, fidelities and eigenstate tests.

Fig. 1 of Faehrmann, Eisert & Kueng, *PRL* **135**, 150603 (2025), reproduced under CC BY 4.0. We use $\phi = -\pi/2$ for the imaginary quadrature, per the paper's body text (below its Eq. 2) and `CONVENTIONS` §5; the printed Fig. 1 caption reads $+\pi/2$.

What a classical shadow actually is

1. Each shot draws its own random basis



$b_j \sim \mathrm{Unif}\{X, Y, Z\}$, independently on every qubit and every shot

All that is stored per shot: $(\, b_0 b_1 b_2,\ \ s_0 s_1 s_2 \,)$

$$\hat{\rho} = \bigotimes_j \left(3\, V_{b_j}^{\dagger} |s_j\rangle\langle s_j| V_{b_j} - \mathbb{1}\right), \qquad \mathbb{E}[\hat{\rho}] = \rho$$

one unbiased snapshot — meaningless alone, exact in expectation once averaged

2. One record, every Pauli at once

$$\hat{P} = 3^{\,w(P)} \prod_{j \in \mathrm{supp}(P)} s_j \, \mathbf{1}[b_j = P_j]$$

shot	basis b	outcome s	$\widehat{Z_0 Z_1}$	$\widehat{X_0 X_1}$
1	Z Z X	+ − +	−9	0
2	X X Z	+ + −	0	+9
3	Z Y Y	− + +	0	0
4	Z Z Z	+ + +	+9	0
5	Y X Z	− − +	0	0

$w = 2$: a shot lands with probability 3^{-2} and the 3^2 weight cancels it exactly, so $\mathbb{E}[\hat{P}] = \mathrm{Tr}[P\rho]$

q_2's basis never enters $\widehat{Z_0 Z_1}$: only $\mathrm{supp}(P)$ must match

$\Rightarrow$ 3^{-w}, not 3^{-n}. The price is $\mathbb{E}[\hat{P}^2] = 3^{w}$.

No new circuits — only ≤ 2 extra single-qubit gates per qubit. The randomisation is the whole trick: it is what makes *one* record set an unbiased estimator for **every** Pauli observable simultaneously, instead of one chosen in advance. [Hsin-Yuan Huang, Kueng & Preskill, *Nat. Phys.* **16**, 1050 (2020)]

One circuit, one set of shots, seven physical quantities

The ancilla alone gives the interference signal:

$$\langle Z_a \rangle_\phi \;=\; \mathrm{Re}\Big[e^{i\phi}\,\mathrm{Tr}(U\rho)\Big]\,, \qquad \phi = 0 \to \mathrm{Re}\,\chi, \quad \phi = -\tfrac{\pi}{2} \to \mathrm{Im}\,\chi$$

Tracing out the ancilla leaves the *unweighted* shadow target, and weighting each snapshot by the ancilla outcome $a = \pm 1$ leaves the *joint* one:

$$\rho^{(I)} = \tfrac{1}{2}(\rho + U\rho U^\dagger), \qquad \mathbb{E}\big[a\,\hat{P}\big] = \mathrm{Re}\Big[e^{i\phi}\mathrm{Tr}(P\,U\rho)\Big]$$

From **one** record set: $\chi(t)$, $\chi_Q(t)$, $\chi_H(t)$, $\chi_{Z_0}(t)$, $\langle H \rangle$, $\langle Q \rangle$, $\langle Z_0 \rangle$ [R009]

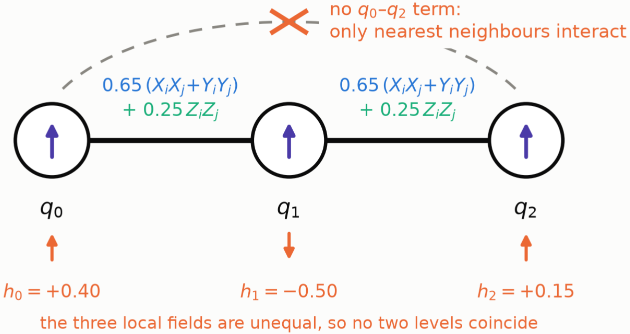
3,456 circuits · **256,000 shots** — one experiment, not seven.

The benchmark: three spins, nearest-neighbour coupling only

3 spin-$\frac{1}{2}$ particles, open chain

conserved: $Q = \sum_j Z_j, \quad [H, Q] = 0$

$\Rightarrow$ the spectrum splits into charge sectors



$$H = \sum_{i=0}^{1} [\,0.65\,(X_iX_{i+1} + Y_iY_{i+1}) + 0.25\,Z_iZ_{i+1}] \;+\; 0.40\,Z_0 - 0.50\,Z_1 + 0.15\,Z_2$$

Why this model. Small enough to diagonalise exactly, so every estimate has a truth reference — yet it still carries both features the protocol exploits: a **conserved charge** and a **non-degenerate spectrum**.

The initial state $R_y^{(q_0)}(1.3)\,|000\rangle$ reaches **4 of the 8 levels**:

$E \in \{-1.9152,\ -0.4810,\ +0.5500,\ +1.9463\}$ in sectors $q = +1, +1, +3, +1$. [R002, R012]

The garbage state

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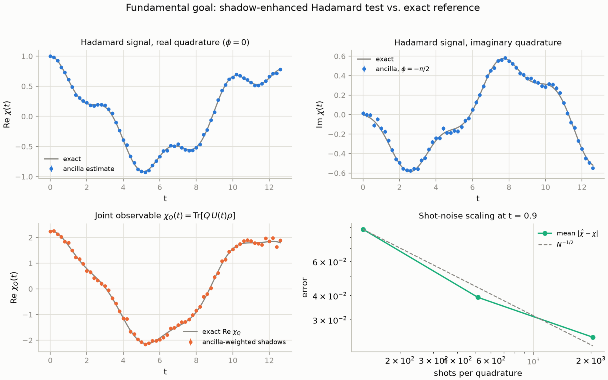
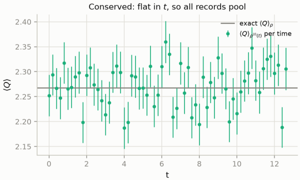
Every estimator agrees with theory — and the error bars are honest

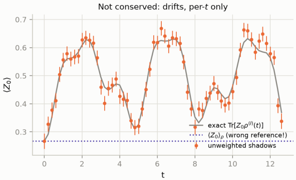
- **56/56 checkpoints pass**, zero hard failures, zero advisory warnings.
- $\chi(t)$ rms error **0.0290** vs predicted sem **0.0277** — ratio **1.05**, so the error bars neither over- nor under-state. [R010]
- Shot-noise slope -0.463 vs theory -0.5: unbiased, statistics-limited. [R010]



The single-shot second moment is exactly $3^{w(P)}$, which predicts the measured error bars to **within 2%**. [R029]

Conserved quantities come out flat — from the very same shots





$$\langle H\rangle = +0.0714 \pm 0.0082$$

exact $+0.0739$

$$\langle Q\rangle = +2.2708 \pm 0.0052$$

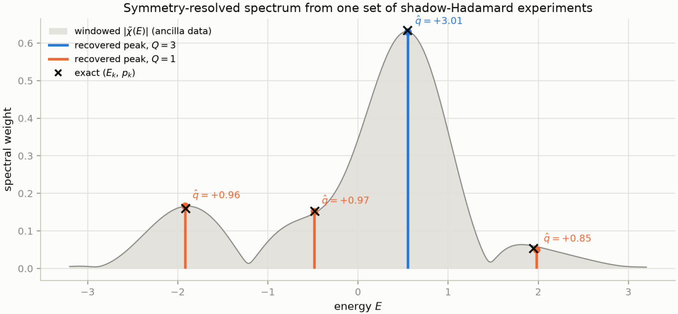
exact $+2.2675$

Pooled over all 128 settings, extracted from the *discarded* register of the $\chi(t)$ experiment. [R010]

Per-time $\langle Q\rangle$ is flat within 5σ at every one of the 64 time points (max 2.2σ). [R010]

No extra circuits (≤ 2 extra 1-qubit gates per system qubit); the statistical cost is on slide 20.

We recover the spectrum *and* label each line by symmetry sector



All 4 populated levels recovered: max $|\Delta E| =$ **0.033**, max $|\Delta p| =$ **0.006**, **every label correct**. [R012, R013]

Reproducible: **12/12 independent seeds** got all four labels right. [R019]

The colour is the deliverable. It is information the grey curve does not contain — and it cost *no additional circuits*.

The grey band is the plain windowed Fourier transform. A better estimator sharpens it; *no* estimator can colour it.

The garbage state

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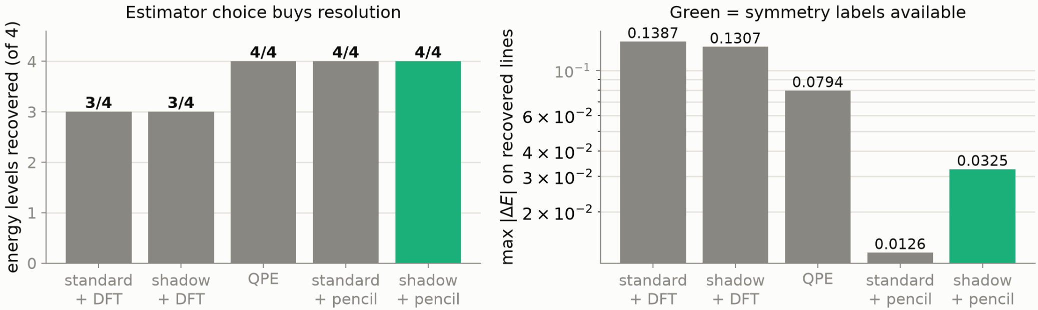
What each choice actually buys

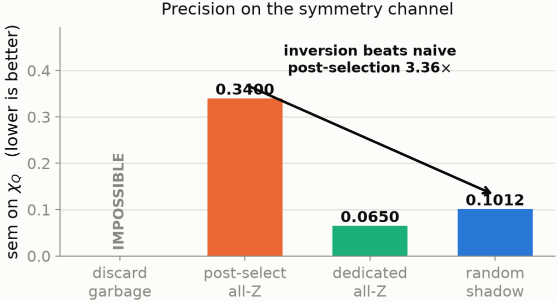
Only the shadow protocol yields symmetry labels — at any shot count



Estimator choice buys resolution (3/4 → 4/4 lines). **Protocol** choice buys the labels — and nothing else can. [R011, R012, R021, R030]

Do you even need shadows? Four ways to treat the garbage

Four ways to treat the garbage register — each rung buys something



What each can measure AT ALL

protocol	$\chi(t)$	Q	H terms
discard garbage	yes	--	--
post-select all-Z	yes	yes	5/9
dedicated all-Z	yes	yes	5/9
random shadow	**yes**	**yes**	**9/9**

Z-only protocols are blind to the 4 hopping terms (0.65 each):
$\langle H \rangle$ is conserved, yet what they report drifts by 0.147
— with no second channel to notice.

Each rung buys a capability the one below *cannot have at any shot count.* [R039, R040]

What the shadow inversion actually contributes

$Q = \sum_j Z_j$ is diagonal in the computational basis, so a **dedicated Z readout beats our shadow estimator by 1.56×** on the symmetry channel. We report that. [R039]

But compare the shadow inversion against *post-selecting the all-Z shots out of its own data*:

$$\underbrace{\tfrac{1}{3}}_{\text{per } \textit{term } Z_j\text{: } b_j = Z} \qquad \text{versus} \qquad \underbrace{\tfrac{1}{3^{\,n}}}_{\textit{all} \text{ qubits Z at once}} \quad = \; 3.7\% \text{ of shots}$$

The inversion beats naive post-selection of the same data by 3.36× (sem $0.3400 \to 0.1012$) — it recovers exactly the shots post-selection discards. [R040]

The case for shadows is not better symmetry resolution.
It is that the *same* shots also give you H — which no Z-only protocol can see.

Which protocol can rebuild which observable?

observable	what it measures physically	**discard** Hadamard → DFT	**post-select all-Z** → matrix pencil	**shadow** → matrix pencil
energies	the level structure of H	3/4 lines	4/4	4/4
Z_i	local magnetisation of spin i	—	yes (err 0.41 at 3.7% yield)	**yes** (0.048)
$Z_i Z_j$	neighbours **aligned** vs **anti-aligned**	—	yes (err 0.28)	**yes** (0.057)
H	total energy	—	**biased** (5/9 terms, 2.70)	**yes** (0.139)
$X_i X_j + Y_i Y_j$	**hopping**: excitation exchanged between i, j	—	**impossible** (0/2 terms)	**yes** (0.509)

A Z-only protocol sees only what is *diagonal*: it can say how the spins are *oriented*, never how they *move*. The hopping term is precisely the part of H that transports an excitation along the chain — the part that makes the dynamics non-trivial — and it converges to $\mathrm{Tr}[O_{Z\text{-diag}} U \rho]$, so more shots do not help.

The shadow's exclusive capability is non-commuting observables. [R041]

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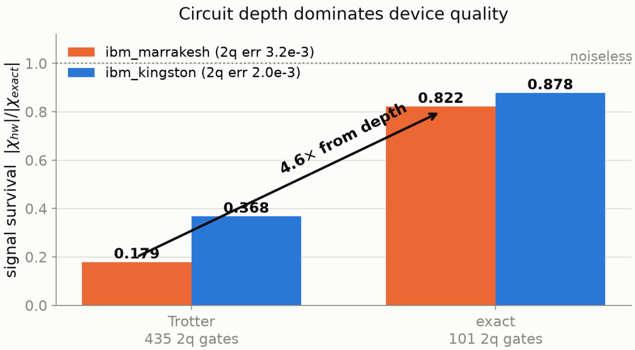
It survives on real hardware once the circuit is shallow enough



Complete 2×2 grid. Depth dominates (**2.4–4.6×**) over device choice (1.1–2.1×). [R008, R018]

With a *full random-basis ensemble* on hardware: χ survival **0.962**, and a valid $\langle Q \rangle$ at **0.957** — the core claim, on a real QPU. [R023]

Caveat, stated plainly: one run per configuration. An independent run of an identical circuit 30 minutes later differed by $> 2\times$. Hardware varies; we report that. [R026, R031]

The honest cost: 13 experiments replaced for a 1.07× variance premium

The per-shot Pauli estimator and its exact second moment:

$$\hat{P} \;=\; 3^{\,w(P)} \prod_{j\in\mathrm{supp}(P)} s_j\, \mathbf{1}[b_j = P_j]\,, \qquad \mathbb{E}\big[\hat{P}^2\big] \;=\; 3^{\,w(P)}$$

Conventional route: **13 dedicated** modified Hadamard tests ($Q{=}3$, $H{=}9$, $Z_0{=}1$ Pauli terms).

Shadow route: **0 extra circuits** (≤ 2 extra 1q gates per qubit).

Price paid instead — measured, not asserted:

$$\frac{\mathrm{sem}_{\text{shadow}}}{\mathrm{sem}_{\text{dedicated}}} = \frac{0.1012}{0.0949} = \mathbf{1.07}\times$$

on χ_Q specifically. [R021]

The 3^w model predicts the observed error bars to **within 2%**. [R029]

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Reproduce it in under five minutes

- **44 sourced results rows** — every number maps to the command that produced it (`EVIDENCE.md`).
- Quickstart reproduces our headline *table* from saved data; the money plot ships pre-rendered.
- `hardware_run.py` reproduces our hardware arm in 3 commands.
- `BUGLOG.md` — including the numbers we got wrong and retracted.

`github.com/mnsh0409/Qiskit-Hackathon-2026`

First open implementation of PRL **135**, 150603. Prior-art search 2026-08-13; scope and limits recorded in `deck/PRIOR_ART_SEARCH.md`.

What we could not do, and where it breaks

- The windowed DFT **never** resolves the 1.031-spaced pair at *any* $T_{\max}$ we tested; the matrix pencil resolves it from $T_{\max} \geq 7.0$. A method limit, not a shot-count limit. [R011, R022]
- Exact-synthesis shallowness is an $n=3$ **artefact** — the advantage falls $3.41\times \to 1.16\times$ from $n=3$ to $n=4$ and is *expected* to invert beyond (measured at $n=3,4$ only). Not a scalable recipe. [R025]
- One noise study came out **confounded**; we exclude it rather than report it. [R017]
- Hardware supports *robustness claims only* — never a headline number. All Part A/B results are simulator.

We make no claim of outperforming classical computation.
The system is 3 qubits; a laptop diagonalises it instantly. It is small *on purpose*, so every estimate can be graded against an exact answer.

Team 8 — Garbage Collectors

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Team 8 — Garbage Collectors

Thank you — questions welcome.

Backup slides

Backup — the standard and shadow protocols give *identical* χ

Measured, not assumed: a genuine textbook Hadamard test over the full 64-point grid at identical budget gives

$$\mathrm{rms}_{\mathrm{std}} = 0.0265, \quad \mathrm{rms}_{\mathrm{shadow}} = 0.0290, \qquad \text{mean sem } 0.0277 \text{ vs } 0.0277$$

with $|\chi_{\mathrm{std}} - \chi_{\mathrm{shadow}}|/\text{combined sem}$: max 2.10σ, mean 0.88σ — statistically the **same random variable**. [R021]

So measuring the garbage register costs the ancilla signal *nothing*, and the two axes factorise cleanly: estimator $\to$ resolution, protocol $\to$ labels.

Backup — a self-validating decoherence witness

For a pure input state, exactly:

$$\mathrm{Tr}\left[(\rho^{(I)})^2\right] = \tfrac{1}{2}\left(1 + |\chi(t)|^2\right)$$

Left side from the **system register alone** (shadow snapshot pairs); right side from the **ancilla alone**. Two independent channels of the same experiment, so their gap certifies decoherence with **no exact diagonalisation and no classical simulation** — it survives to sizes where nothing can be verified.

ideal simulator	-0.1σ	(correct null)
noisy sim, $1\times$	-7.6σ	(fires)
noisy sim, $5\times$	-17.2σ	(dose-response)
real hardware (cleanest run)	-1.4σ	(correct null)

[R033]

Backup — future work, already prototyped: density of states

The DOS is the Fourier transform of the *trace* of the propagator:

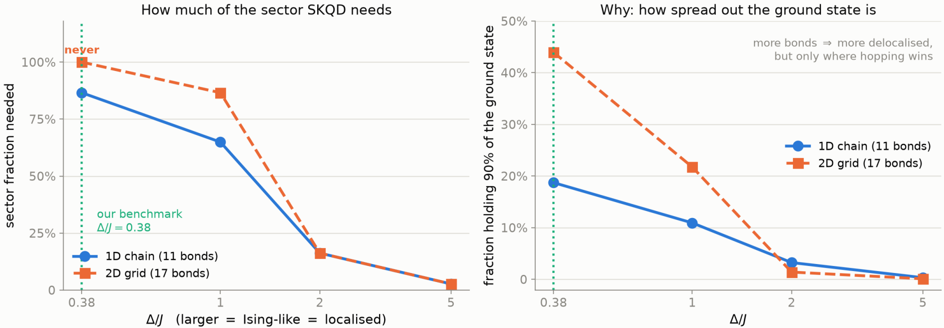
$$G(t) \;=\; \frac{1}{\sqrt{2\pi}}\,\mathrm{Tr}\Big[e^{-i\hat{H}t}\Big] \qquad \text{(Goh \& Koczor, arXiv:2407.03414v2, Eq. 7)}$$

Our $\chi(t) = \mathrm{Tr}[U(t)\rho]$. Set $\rho = \mathbb{1}/2^n$ and **the same experiment becomes a DOS estimator** — one line changed, analysis chain untouched.

Prototyped: peak weights correctly become *degeneracies* (isolated levels 0.11–0.13 vs predicted 1/8; a 3-fold merged peak 0.354 vs predicted 3/8). [R038]

Where we could improve on that paper: it obtains fixed-particle-number DOS with *one experiment per sector*. Our shadow channel measures $\langle Q^k \rangle$ from the *same* shots, so the charge moments at each peak recover **all sectors from a single run**. Feasibility checked: Q^2, Q^3 are 4 Pauli terms each; moment inversion condition number 31.

Backup — where SKQD helps, and why not here



Sample-based Krylov diagonalisation spans a subspace instead of enumerating one. Its advantage is governed by **how localised the ground state is**, which Δ/J controls: at $\Delta/J = 5$ it needs **2.7%** of the charge sector, at our benchmark's $\Delta/J = 0.38$ it needs **86.6%** — so SKQD buys us nothing *on this Hamiltonian*. Measured at 12 sites, sector $\binom{12}{6} = 924$, against exact diagonalisation. [R047]

And in 2D? Same 12 sites and same sector, 11 bonds $\rightarrow$ 17. More connectivity makes it **strictly worse where hopping dominates** (2D never converges at $\Delta/J = 0.38$) and **changes nothing once localised** ($\Delta/J \geq 2$: identical in both). A 2D version of our benchmark would be worse for SKQD, not better. [R048]

Backup — two bugs we caught, and how

B04 — a fixed-basis shadow estimator silently returns a *different* observable. The estimator is unbiased only for i.i.d.-uniform bases; with one fixed basis the indicator is deterministic and non-matching terms contribute structurally zero. We reported a hardware $\langle Q \rangle$ this way and blamed noise. On a *noiseless* simulator the same code is **65σ off** — so it was never noise. Retracted; `hardware_run.py` now refuses to print system observables without `--shadow`.

B06 — an error bar computed over the wrong units. Purity is a U-statistic over *pairs* of snapshots; we used $\mathrm{std}/\sqrt{n_{\mathsf{pairs}}}$, but pairs share shots, so the effective sample size is the number of *shots*. That faked a 2.2σ signal. **Only the ideal-simulator control leg exposed it** — the hardware number alone looked fine.

A witness validated only where the truth is unknown cannot be trusted.